

Call for Expressions of Interest

<i>Date of Issue:</i>	September 2026
<i>Expected Completion Date:</i>	May 2027
<i>Project Title:</i>	Feeling the Music
<i>Company:</i>	UoN/ VIP Studio

Contact Details

<i>Lead Contact:</i>	Paul Tennent
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Additional Info

<i>Students must sign NDA:</i>	☐
<i>Project uses personal data:</i>	☐
<i>Custom hardware required:</i>	☒ If Applicable: Tesla Suit (Provided by VIP Studio)
<i>Relevant research areas (if applicable)¹:</i>	Human Computer Interaction

Project Background Information

Full body haptics are a bit of a dream for working and playing in virtual reality.

This project invites you to make use of our Teslasuit system – a full body electro-muscle stimulation system which can create a range of sensations across the body. In this project you will explore the notion of synaesthesia. That is, you will create a system where musical performance is transformed from sound into tactile sensation. The team will develop a system using the Teslasuit to covert sound to touch, analysing the character of the sound in real time and creating appropriate sensations through the suit for a listener to experience sound and music in a new way. Extending beyond this may include the development of visualisations of the sound delivered through mixed reality, and even exploring what it feels like to play an instrument when you can feel the sound on your skin.

For more information on the suit itself, see here: <https://teslasuit.io/>

¹ An overview of different research areas in the school of computer science can be found [here](#).

Requirements	
<i>The group must provide:</i> The group will develop a program, running on a network-connected computer that 1) listens to audio through a microphone 2) analyses the characteristics of that audio 3) generates appropriate sensation through the Teslasuit 4) (optional) generates visualisation of the audio through a VR headset The group will also test their system with real humans and reflect on what it feels like to experience music in this way. It will be necessary to review existing work on the actuation and visualisation of music. The group will also have to develop a safe and appropriate onboarding and offboarding process for working with the teslasuit. Note we have a policy that nobody works with the teslasuit alone, so you should be prepared to work together for any and all development when the suit is involved.	
Skills	
<i>Highly desirable:</i> • Interest in music and or music theory. • Ability to work together on a creative but technically demanding project. • Strong communication skills. • Ability to program in C++, C-Sharp and possibly python. • At least one of you an average physical size (the suits are quite specifically sized and we only have medium)	<i>Desirable:</i> • Some experience with the Unity platform. • Some knowledge of audio signal processing.
Submission / Award information	
<i>Date of Submission:</i>	September 2026
<i>Date of Pitch:</i>	October 2026
<i>Notification of award:</i>	October 2026

For further information on this proposal, the team representative should in the first instance e-mail the lead contact stated above at the e-mail address provided.